ShelfSense AI: Autonomous Retail Compliance Agent

The Challenge

- Retail stores face operational and revenue losses due to misplaced products, incorrect shelf labels and prices, stock-outs, and deviations from planned shelf layouts. Manual shelf audits are time-consuming, error-prone, and difficult to scale.
- Build an AI-powered retail compliance system that analyses shelf images or videos to automatically detect merchandising and compliance issues, helping store managers maintain accurate, well-organized, and adequately stocked shelves.

The Solution Should Be Able To

- **Validate product placement and planogram compliance:** Detect shelf products, compare their placement and arrangement against the expected planogram, identify missing or misplaced products, recommend correct locations, and generate a compliance score.
- **Verify product labels and prices:** Match products with their corresponding shelf labels, extract displayed prices using OCR, compare them against a reference product catalogue, and flag product-label or pricing mismatches.
- **Detect shelf availability issues:** Identify empty shelf spaces, low-stock situations, and products requiring replenishment.

Expected Output

- The solution should generate a clear report or dashboard highlighting detected violations, shelf availability issues, recommended corrective actions, and an overall planogram compliance score.
- Additional points will be awarded to teams that successfully deploy or run inference on an edge device or in an edge-simulated environment.

Auto-Settle AI: Autonomous Video-to-Cash Claims Adjuster

The Challenge

- Minor motor-insurance claims, such as dents, scratches, and broken mirrors, can take several days to process due to manual vehicle inspections, repair-cost estimation, and settlement discussions.
- Build a multimodal AI-powered claims-adjustment system that analyses images or videos of vehicle damage, estimates the repair cost, and interacts with the customer to arrive at an instant cash-in-lieu settlement within predefined financial and operational guardrails.

The Solution Should Be Able To

- **Assess vehicle damage:** Analyse uploaded images or videos to identify the damaged vehicle component, classify the type of damage, and estimate its severity.
- **Estimate repair and settlement costs:** Map the damage assessment to a reference database containing parts and labour costs, calculate the estimated garage repair cost, and establish an allowable cash-settlement range.
- **Facilitate an automated settlement:** Present an initial cash offer, respond intelligently to customer counteroffers, and arrive at a final settlement without exceeding the predefined financial limits.
- **Apply guardrails and escalation rules:** Prevent settlements outside the approved range, identify uncertain or suspicious claims, and escalate the case to a human claims adjuster when an automated decision cannot be made safely.

Expected Output

The solution should provide a **website or application** that accepts vehicle images or videos, presents the detected damage and estimated repair cost, conducts an interactive settlement conversation, and produces either a final cash offer or a human-escalation decision with a clear audit trail.

Fresh-Guard AI: Perishable Inventory and Merchandising Agent

The Challenge

- Grocery stores face significant losses when perishable products remain unsold, become overstocked, or deteriorate before corrective action is taken. Manual inspection of fresh produce and dairy shelves is time-consuming and often reactive.
- Build an AI-powered system that analyses images or videos of perishable inventory, evaluates product freshness and stock levels, and recommends timely pricing, merchandising, and promotional actions to reduce food waste while protecting store margins.

The Solution Should Be Able To

- **Monitor freshness and inventory:** Analyse shelf or produce-bin images and videos to classify product freshness, ripeness, or deterioration, while estimating stock levels and identifying overstocked items.
- **Recommend pricing and merchandising actions:** Suggest suitable markdown prices for deteriorating products while considering cost and margin constraints. For fresh but overstocked products, recommend bundles, product placements, or cross-merchandising strategies to improve full-price sales.
- **Generate targeted promotions:** After the store manager approves an action, generate contextual promotional content for a mock customer loyalty platform, including relevant discounts, bundles, recipes, or product-use suggestions.

Expected Output

The solution should provide a website or application that displays detected freshness and inventory issues, recommends pricing or merchandising actions for managerial approval, and generates customer-facing promotional content for the approved action.

CareFlow AI: Human-in-the-Loop Healthcare Triage Assistant

The Challenge

- Hospitals and clinics often receive patient information in many different forms, such as typed symptom descriptions, voice recordings, medical reports, and images. The information may be incomplete, unclear, or provided in different languages. High patient volumes and limited availability of healthcare professionals can also delay review.
- Build a **multilingual, multimodal AI assistant** that converts this information into a clear, structured summary and helps healthcare professionals decide **which patients may need attention first**.
- The AI should **not diagnose diseases or make the final medical decision**. It should identify possible urgency signals, clearly show when it is uncertain, and ensure that a qualified healthcare professional makes the final decision.

The Solution Should Be Able To

- **Capture and structure patient information:** Accept symptoms through text or voice, extract useful information from sample medical reports using OCR, organize when symptoms started and how they changed, and identify important information that may be missing.
- **Support multilingual interactions:** Understand and translate inputs in English, Hindi, regional Indian languages, and mixed-language conversations such as Hindi-English. Clearly indicate when the system is unsure about a transcription or translation.
- **Provide safety-first triage support:** Use simple rules and/or AI to identify possible warning signs, prioritize cases for review, suggest useful follow-up questions, and immediately flag potentially urgent cases for a healthcare professional.
- **Handle uncertain or conflicting information:** Detect and highlight cases where information does not match, a report is difficult to read, extracted information has low confidence, or an input contains suspicious instructions. The system should **flag these issues rather than guess or silently correct them**.
- **Enable human review and referral:** Provide healthcare professionals with a prioritized list of cases. Show where important information and warning flags came from, and allow reviewers to approve, edit, or override AI-generated notes before they are finalized.
- **Protect patient information:** Ask for patient consent, allow patient data to be withdrawn or deleted, restrict information based on user roles, and maintain a record of important actions performed by both the AI system and human reviewers.

Expected Output

The solution should be a **website or application** that demonstrates:

- A prioritized dashboard showing which cases should be reviewed first
- A structured summary of each patient's symptoms and information
- Clear alerts for potentially urgent cases
- Flags for missing, unclear, or conflicting information
- AI-suggested follow-up questions
- Links between extracted information and its original source, such as a report, voice recording, or patient message
- A final referral or triage note that can be **reviewed, edited, and approved by a healthcare professional**

Common Ask

Before building the solution, teams are expected to **research and understand the background of the given use case**, including the real-world context, users, workflows, and key challenges involved. Since problem statements may not specify every detail, teams should make **reasonable and valid assumptions** wherever required and clearly state these assumptions as part of their solution. Teams are encouraged to think beyond the requirements provided, but any additional features, data, workflows, or assumptions should be **realistic, relevant, and supported by their research**.

Bonus Considerations

Additional points may be awarded for:

- **Efficient SLM Usage:** Teams may use an **LLM of any size**. However, additional points may be awarded for effectively using a Small Language Model with fewer than 20 billion parameters, where appropriate, while maintaining strong performance and reducing latency or resource requirements.
- **Additional Innovation:** Beyond the capabilities specified in each challenge, teams that introduce relevant additional features, workflows, or domain-specific innovations that improve the solution's usefulness, robustness, or real-world applicability may receive additional points.